

Printer Bridge Setup Guide

⚠ Important Architecture Note

You requested to deploy this bridge to **Vercel**. However, **this is not possible** for Direct LAN Printing, and here is why:

- **Vercel** runs in the "Cloud" (Internet).
- **Your Printer** is on your "Local Network" (LAN) at 192.168.10.15.
- Cloud servers **cannot** connect to your private Local IPs.

The Solution:

The **Frontend website** lives on Vercel.

The **Printer Bridge** lives on your **Local Computer** (the POS terminal or laptop in the restaurant).

The Frontend talks to the Bridge via localhost (Internal Device Communication), and the Bridge talks to the Printer.

[Frontend (Vercel)]	--->	[Browser (Your PC)]	--->	[Bridge (Your PC)]	--->	[Printer (LAN)]
(Internet)		(Localhost)		(127.0.0.1)		(192.168.10.15)

Setup Instructions

1. Prerequisites

- The computer connected to the printer (or on the same Wi-Fi) must have **Node.js** installed.
 - Download: <https://nodejs.org/> (<https://nodejs.org/>) (LTS Version)

2. Installation (One Time)

1. Open your terminal or command prompt.
2. Navigate to the project folder:

```
cd "User Files/ASU/Internship/QR Code again/printer-bridge"
```

3. Install the required dependencies:

```
npm install
```

3. Running the Bridge

Whenever you open your restaurant, you must start the bridge.

Option A: Manual Start

1. Open Terminal.
2. Run:

```
npm start
```

3. Keep this window open. You will see: Printer Bridge Server running on <http://localhost:3001>

Option B: Auto-Start (Recommended)

To make it start automatically when the computer turns on, we will use the **Windows Startup Folder**. This is easier and more reliable than PM2 on Windows.

1. **Right-click** the start_bridge.bat file you just created.
2. Choose **Create Shortcut** (it will create a file like start_bridge - Shortcut).
3. Press **Windows Key + R** on your keyboard.
4. Type shell:startup and press **Enter**.
 - *This opens the folder where Windows looks for programs to start automatically.*
5. **Drag and drop** the shortcut you created into this folder.

That's it! Next time you restart the computer, the Printer Bridge will open automatically.

Verifying It Works

1. Ensure the bridge is running.
2. Open your Vercel-hosted App.
3. Go to **Admin > Printer Management**.
4. Click **"Add Printer"**.
 - **IP:** 192.168.10.15 (Your specific printer IP)
 - **Port:** 9100 (Default)
 - **Type:** Kitchen
5. Click **Save**.
6. Click **"Test Print"**.
 - You should see a log in your terminal: Connecting to printer... Data sent.
 - The printer should print a small test ticket.

Troubleshooting

- **"Direct print failed"**: Check if the black terminal window with npm start is running.
- **"Printer connection timeout"**: Check if your computer is on the same Wi-Fi/Ethernet as the printer. Ping the printer IP from your terminal: ping 192.168.10.15
- **Windows Error: "npm.ps1 cannot be loaded because running scripts is disabled"**:
 - This is a Windows security setting in PowerShell.
 - **Fix:** Run this command in PowerShell:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

- Press Y when asked. Then try npm install again.
- *Alternative:* Use **Command Prompt (cmd)** instead of PowerShell.

Moving to Another Computer

If you want to run this bridge on a different computer (e.g., the dedicated POS terminal):

1. **Copy the Folder:** You can copy the entire printer-bridge folder to the new computer.
 - *Tip: You don't need to copy the node_modules folder. It's cleaner to delete it before*

copying and reinstall it on the new machine.

2. **Install Node.js:** The new computer **MUST** have Node.js installed.

- Download: <https://nodejs.org/> (<https://nodejs.org/>)

3. **Install Dependencies:**

- Open terminal in the new folder.
- Run npm install (this recreates the node_modules folder).

4. **Start the Bridge:** Run npm start.